Module 10 – Reflective Journal

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This lab helped me understand unsupervised learning and how machine learning can find patterns without using labeled data. I learned that supervised learning uses known answers during training, while unsupervised learning focuses on finding similarities and patterns. Before this lab, I did not fully understand how a model could organize data without knowing the correct answers.

One important part of the lab was K-Means clustering. I learned how the elbow method helps determine a suitable number of clusters. The graphs made it easier to see how data points were grouped and why choosing the right number of clusters matters. I also learned about Principal Component Analysis (PCA), which reduces the number of features while keeping most of the important information. Using the Iris dataset helped me understand how PCA makes complex data easier to visualize.

Overall, this lab helped me understand that machine learning is not just about running code and getting results. Each step requires understanding the data, choosing the right method, and interpreting the output. I also realized that visualization is important because it helps explain results that might be difficult to understand from numbers alone. These concepts could be useful in real-world applications such as customer segmentation and data analysis.

Steven Ford's Reflection

Working through unsupervised learning highlights a fundamental shift in analytical mindset: rather than optimizing a model to predict an external ground-truth target, the focus turns to evaluating structural integrity, information density, and geometry within the feature space itself. The primary conceptual challenge shifts from tracking empirical loss curves to diagnosing trade-offs between compression and granularity, balancing parsimonious cluster boundaries against the risk of over-segmentation. Concurrently, analyzing variance retention demonstrates how latent representations expose underlying collinearity, showing that high-dimensional complexity can frequently be distilled into a far lower intrinsic dimensionality without sacrificing essential separability. Ultimately, this exercise underscores that unsupervised methods serve less as automated predictors and more as critical diagnostic instruments—revealing natural distributions, eliminating redundancy, and providing rigorous geometric validation prior to downstream modeling.

Antonio Garcia

In this lab I worked with unsupervised learning and learned how machine learning can find patterns in data without using labels. I used two main techniques which were K Means clustering and Principal Component Analysis. Both helped me understand how data can be grouped or simplified in ways that make it easier to analyze. The first part of the lab used synthetic customer data created with make_blobs. Even though the data was artificial, it helped me see how clustering works. When I plotted the points I could see that the data naturally formed groups. Using K Means made this structure clearer. I learned how inertia measures how tightly the data points fit within their clusters and how the elbow method helps choose a good value for k. I also learned that inertia always decreases as k increases, but choosing a very large k does not give meaningful clusters. The elbow point showed where adding more clusters stopped giving useful improvement.

The PCA section taught me how dimensionality reduction works. The Iris dataset has four features, but PCA reduced it to two principal components while still keeping more than ninety five percent of the variance. This showed me that the dataset has a lot of structure and redundancy. Plotting the PCA results made it easy to see how the species separate in the new two dimensional space. I also learned how PCA can make data easier to visualize and faster to process, especially when dealing with high dimensional datasets. One challenge I faced was that some of my code cells became compressed into a single line when I pasted them into Colab. This caused syntax errors and missing imports. I had to fix the formatting, restore line breaks, and re add libraries like seaborn and matplotlib. Once the cells were repaired everything ran correctly. This reminded me how important clean formatting is in Python, especially when working with loops, comments, and multiple imports.

Overall this lab helped me understand how unsupervised learning can reveal patterns that are not obvious at first. Clustering can group similar data points and PCA can simplify complex datasets without losing important information. I can see how these techniques apply to cybersecurity. Clustering could help group similar alerts or detect unusual behavior, and PCA could reduce noisy high dimensional security data to something more manageable. This lab made unsupervised learning feel more practical and connected to real problems, and it helped me build confidence in working with these tools.

Cristian Gamez Delagdo

In the notebook, we went a little more in depth for the concepts of supervised and unsupervised learning, specifically some of the process involved in unsupervised learning. Unsupervised learning involves datasets with no labels, relying on the algorithms themselves to find features and figure out patterns in the data. The 2 techniques described in the notebook are K-means clustering (used for finding groups of data)and Principal Component Analysis to reduce dimensionality. The Elbow Method is a means of identifying the ideal number of clusters used in K-Means Clustering. Where the elbow curve starts to abruptly flatten out is where the ideal K number is revealed. PCA is meant to simplify more complex datasets while still retaining the more important features. In the example given, the iris dataset is discovered to have 2 main features that dictate the direction of the data, and thus this is what is featured in presented in the graph. One question I have is if there are other techniques that can reduce the dimensionality besides PCA.